

KARAN CHAWLA

karan.chawlad@gmail.com [✉](#) | +1 613 462 4971 | [LinkedIn](#) [✉](#) | [GitHub](#) [✉](#) | [Portfolio](#) [✉](#)

EXPERIENCE

UofTHacks

Oct. 2025 – Present

Full Stack Developer

Toronto, ON

- Architecting high-traffic frontend interfaces using **React** and **Next.js** to support **3,000+** concurrent users during the 2026 registration cycle.
- Optimized data fetching patterns between client and REST APIs, ensuring sub-second page load times for the hacker dashboard.
- Engineered an automated support chatbot on Hacker Dashboard using **OpenAI API**.

SAE AutoDrive - Toronto Autonomous Vehicle Team

Sep. 2025 – Present

State Estimation Developer (C++/Python)

Toronto, ON

- Developing a **lane localization** and state rollback algorithm in **C++**, improving positional accuracy by 15% for Level 4 autonomous maneuvers.
- Implementing **ROS**-based diagnostic tools to analyze sensor fusion pipelines, identifying and resolving critical edge-case failures in GPS/IMU synchronization.
- Collaborating via **Git** and Agile methodologies to integrate estimation modules into the vehicle's real-time control stack.

PROJECTS

LipsLipsRevolution — HackTheNorth Winner | *Next.js, WebSockets, MongoDB, FastAPI*

Sep. 2024

- Engineered a real-time full-stack game that translates lip movement into text via the SymphonicLabs API.
- Reduced data transmission latency by **76% (30s to 7s)** by reverse-engineering proprietary endpoints and implementing their **WebSocket API**.
- Designed a scalable leaderboard system handling live concurrent updates for 50+ players using an event-driven architecture.

Echo - Agentic Voice AI to Get Help | *React, Express.js, VAPI, Twilio*

Aug. 2025

- Built an end-to-end **Full Stack** application to automate outbound phone communications using autonomous voice agents.
- Integrated **Twilio** and **VAPI** to handle real-time audio streaming, transcription, and LLM-based decision-making during active calls.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, Java, HTML/CSS, SQL

Frameworks: React, Next.js, Express.js, Node.js, Flask, TailwindCSS, ROS

Tools & Cloud: Git/GitHub, Docker, WebSockets, GCP, Firebase, Supabase, Vercel, MongoDB

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science — Engineering Science

Sep. 2025 – April 2030

- Focus:** Robotics and Software. Curriculum in calculus, linear algebra, and computer science fundamentals.

International Baccalaureate Diploma

Ottawa, ON

Merivale High School — Score: 41/45 (97.7% Average)

June 2025